PicoDB: Lightweight File-Based Database System

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What is PicoDB?

- Lightweight database built in C++
- File-based storage system
- Supports basic SQL operations (CREATE, INSERT, SELECT, UPDATE, DELETE)
- Includes indexing and networking capabilities

Why PicoDB?

The Challenge

Traditional DBMS are complex and heavy

Difficult to understand internal working

The Goal

Learn database internals hands-on

Provide simple and efficient storage system

How it Works

File Storage

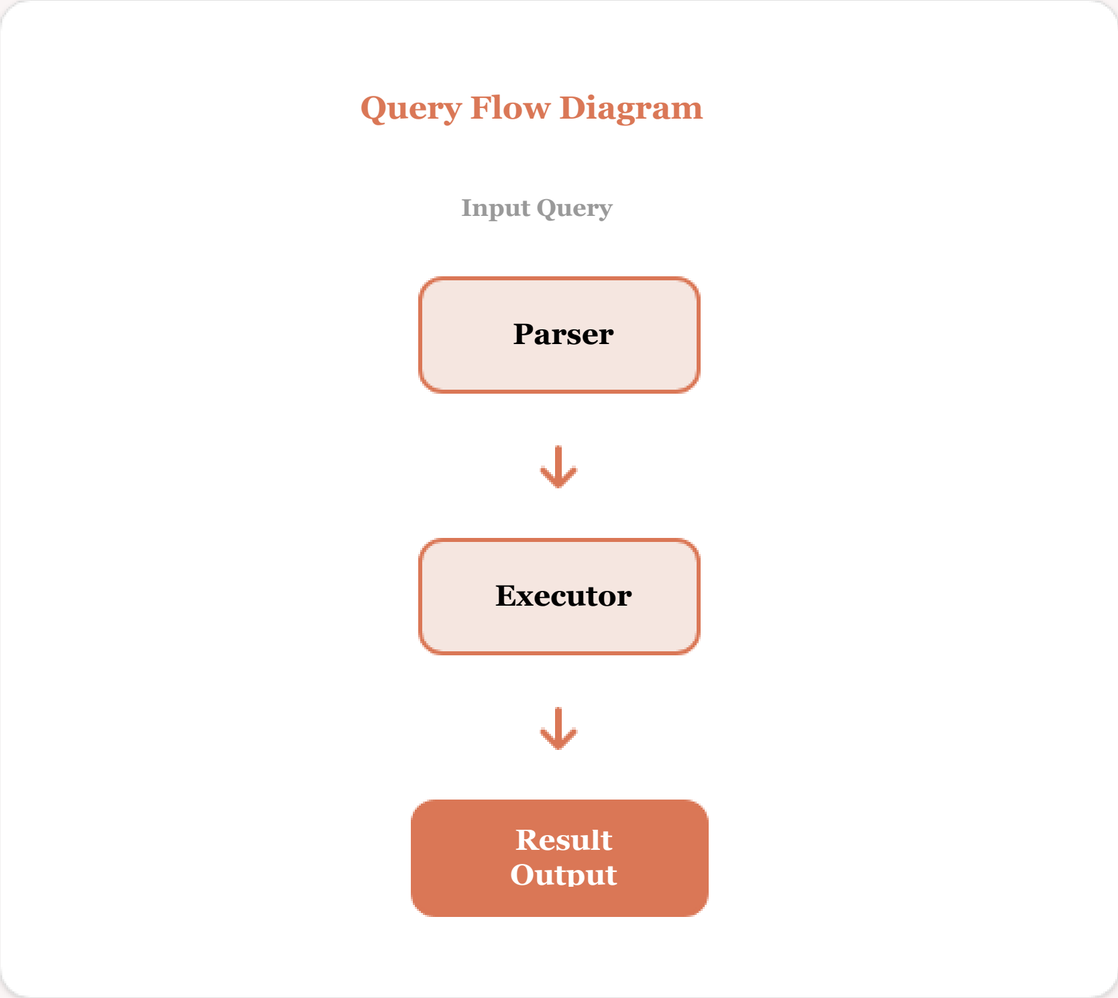
Data stored in files (.data, .meta)

Indexing

Hash index for fast lookup, B+ Tree for range queries

Query Flow

Parse → Execute → Return result



Key Features

CRUD Operations

INSERT, SELECT, UPDATE, DELETE operations

Update Strategy

In-place update + Copy-on-Grow strategy

Soft Delete

Tombstone (soft delete) mechanism

Index Persistence

Index persistence across sessions

Networking

Multi-client support via TCP

Advanced Feature: B+ Tree

Efficient range queries

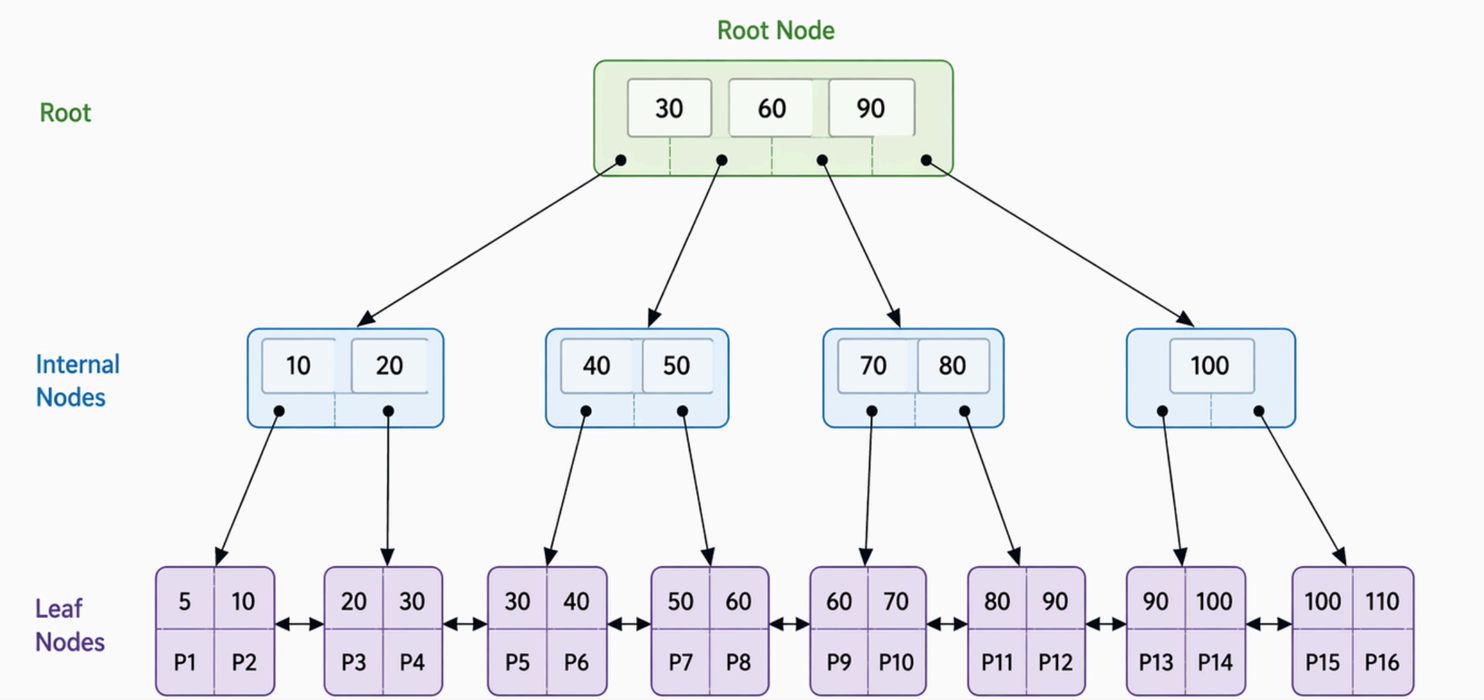
(BETWEEN operations)

O(log n) performance

for lookups

Balanced tree structure

for optimal access



Socket Programming

Client-server model

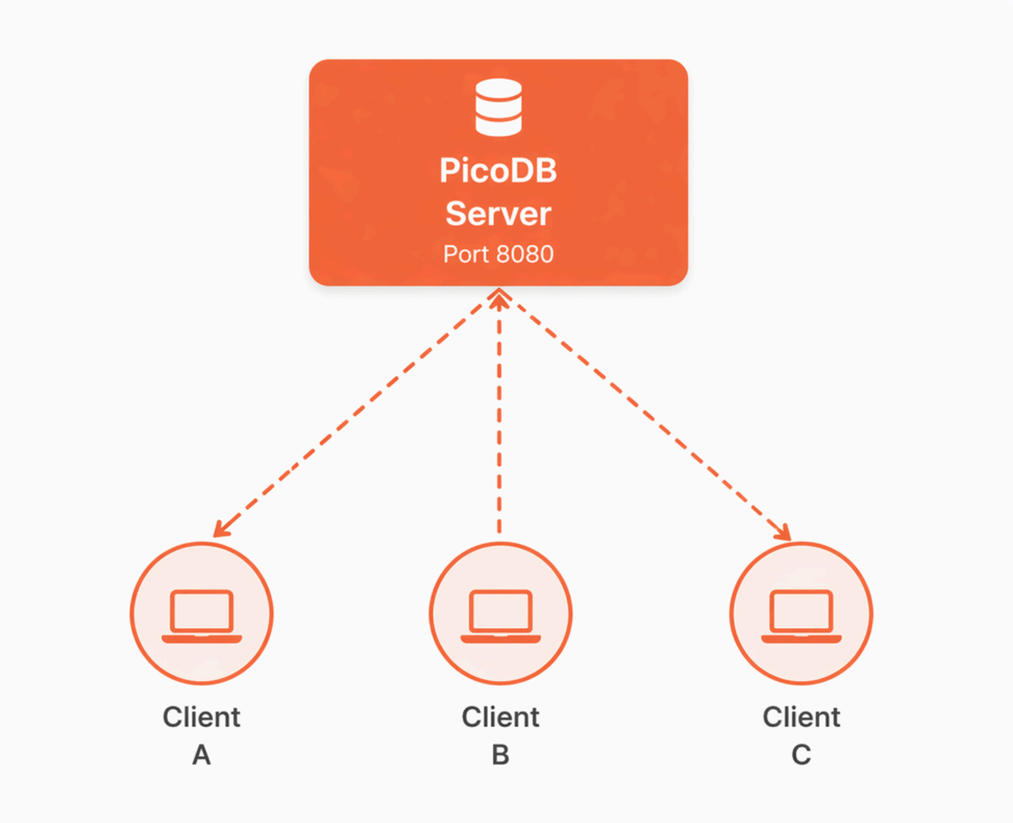
Robust architecture for handling database requests via TCP sockets.

Multiple users can query simultaneously

Concurrency support using pthreads to manage multiple client connections.

Remote database access

Access PicoDB from any remote machine on the network via port 8080.



Conclusion

- Fully functional file-based database system
- Supports storage, indexing, queries, and networking
- Built from scratch in C++